

# Lecture 01

## Computer Systems

Chapter lecture — 12 topics with explanations, worked examples, Urdu notes, practice questions and a full answer key.

BIEK Computer Science XI · Bilingual support: English + اردو

بی آئی ای کے کمپیوٹر سائنس — گیارہویں جماعت

12 book lectures in this pack

★ Golden topics, Urdu notes, review and answer key

XI

# How to use this BIEK study-guide lecture

Each PDF follows one textbook lecture from the BIEK / Sindh Computer Science notes (New Sindh Curriculum). Read *Learn it*, study the SVG diagram, try the worked example on paper, then answer *Check yourself* before the answers. **Extra board practice** (MCQs, short and long questions) is added after the book content — it is not in the textbook. Career-only extras that were not in the book lecture list have been removed. Math is rendered with MathJax. ★ Golden topics are high-yield in board exams. The green box repeats the idea in Urdu.

ہر لیکچر کتاب کے درس کے مطابق ہے۔ کتاب کے بعد اضافی MCQs اور سوالات ہیں۔ جو اضافی کیئریر صفحات کتاب کی فہرست میں نہیں تھے وہ ہٹا دیے گئے ہیں۔

## Lectures in this pack

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Chapter 1: Computer Systems. Study one lecture at a time, then use the chapter review and answer key at the end of this PDF.

**L01.** 1.1.1 & 1.1.2 Discrete vs Continuous & Digital Systems

**L02.** 1.1.3 Analog and Digital Signals

**L03.** 1.1.4 – 1.1.6 Boolean Algebra, Operations & Truth Tables

**L04.** 1.1.7 Basic Logic Gates (AND, OR, NOT)

**L05.** 1.1.7 Universal & Advanced Logic Gates

**L06.** 1.1.8 – 1.1.11 Logic Expressions, Minterms & Diagrams

**L07.** 1.1.13 Karnaugh Maps (K-Maps) Simplification

**L08.** LogiSim Simulation Tool Guide

**L09.** 1.2.1 – 1.2.3 Software Development Life Cycle (SDLC)

**L10.** 1.2.4 – 1.2.7 Waterfall vs Agile Models & Case Studies

**L11.** 1.3.1 – 1.3.4 Communication Models & OSI 7-Layer Model

**L12.** 1.3.5 – 1.3.6 TCP/IP Model vs OSI Model

# Computer Systems

This chapter shows you how a computer thinks in 0s and 1s, how logic gates turn those bits into decisions, how software is planned and built step by step, and how data travels across a network. These three ideas — digital logic, software development and communication models — are the foundation of everything else in Computer Science.

اس باب میں آپ سیکھیں گے کہ کمپیوٹر صفر اور ایک کی زبان میں کیسے کام کرتا ہے، لاجک گیٹس فیصلے کیسے کرتے ہیں، سافٹ ویئر مرحلہ وار کیسے بنایا جاتا ہے، اور نیٹ ورک پر ڈیٹا ایک جگہ سے دوسری جگہ کیسے پہنچتا ہے۔

## What you will learn

- Tell the difference between discrete and continuous quantities, and between analog and digital signals, with examples.
- Use the Boolean operations AND, OR and NOT, and apply the order of precedence.
- Draw the symbol, write the expression and give the truth table of AND, OR, NOT, NAND, NOR, XOR and XNOR gates.
- Simplify Boolean expressions using 2-variable and 3-variable Karnaugh maps.
- Describe the six phases of the SDLC and compare the Waterfall and Agile models using case studies.
- State the advantages of digital systems.
- Build a complete truth table for any expression with 2 or 3 inputs.
- Write minterms (SOP) and maxterms (POS) from a truth table, and draw a logic diagram from an expression.
- Build and test a simple circuit in LogiSim Evolution.
- Explain the seven layers of the OSI model and map them to the four layers of the TCP/IP model.

1.1.1 & 1.1.2 Discrete vs Continuous & Digital Systems

LEARN IT

Everything around you that can be measured or counted is a **quantity**. Quantities come in two kinds.  
**Discrete quantities** are separate, countable items with specific values. You can count them one by one: 1, 2, 3 ... There is nothing "in between". You cannot have 2.5 students in a classroom.  
**Continuous quantities** can take any value within a range. They flow smoothly without breaks. Water can be 30 °C, 30.5 °C, 30.51 °C, and so on — there is always another value in between.

| Discrete (counted)                | Continuous (measured)    |
|-----------------------------------|--------------------------|
| Number of students in a classroom | Temperature of water     |
| Number of cars in a parking lot   | Speed of a moving car    |
| Has fixed, separate values        | Has any value in a range |



Stairs have fixed steps (discrete). A ramp lets you stand at any point (continuous).

REMEMBER IT

Stairs vs Ramp. On stairs you can only stand on step 1, 2 or 3 — that is discrete data. On a ramp you can stand at any height — that is continuous data. Ask yourself: "Do I count it or measure it?" Count → discrete. Measure → continuous.

COMMON MISTAKE

Students sometimes say "money is continuous because amounts are large". No — rupees and paise are counted in fixed units, so money is discrete. Size does not matter; what matters is whether values in between are possible.

آرڈو میں سمجھیں

منفصل (Discrete) مقداریں وہ ہیں جنہیں ایک ایک کرکے گنا جا سکتے، جیسے کلاس میں طلبہ کی تعداد۔ مسلسل (Continuous) مقداریں کسی حد تک اندر کوئی بھی قیمت لے سکتی ہیں، جیسے پانی کا درجہ حرارت۔ سبزہاں منفصل ڈیٹا کی مثال ہیں اور لہلوں (ریزپ) مسلسل ڈیٹا کی۔

CHECK YOURSELF

1. Is the number of pages in a book discrete or continuous?  
2. Is your height discrete or continuous?  
3. In the stairs/ramp analogy, which one shows continuous data?

Answers: 1. Discrete (you count pages). 2. Continuous (height is measured and can be any value). 3. The ramp.

LEARN IT

A **digital system** is an electronic system that uses only two values: 0 and 1. These two values are called **binary digits**, or **bits** for short.

- 0 represents OFF, LOW, or FALSE.
- 1 represents ON, HIGH, or TRUE.

Why only two values? Because it is easy for an electronic circuit to tell "voltage present" from "no voltage". Every number, letter, photo and song in a computer is finally stored as a long pattern of these bits.

| Advantages of digital systems |                                          |
|-------------------------------|------------------------------------------|
| Advantage                     | Detail                                   |
| Reliable                      | Less affected by noise and interference. |
| Accurate                      | No loss of quality when copying data.    |
| Easy to design                | Only two states (0 and 1) to handle.     |
| Programmable                  | Can be easily controlled by software.    |

**Examples of digital systems**  
**Computers:** process data as 0s and 1s.  
**Smartphones:** store photos and music as binary data.  
**Digital cameras:** capture images as binary code.

WORKED EXAMPLE

Why is copying digital data "accurate"?

Step 1: A song is stored as bits, for example 10110010..  
Step 2: When you copy it to a USB, the computer copies each bit exactly: 1 stays 1, 0 stays 0.  
Step 3: Even after 100 copies the bits are identical, so the song sounds the same. An old cassette tape (analog) loses a little quality every time it is copied.

REMEMBER IT

Think of a bit as a light switch: it is either ON (1) or OFF (0). A digital system is a huge panel of such switches.

آرڈو میں سمجھیں

ڈیجیٹل نظام صرف دو قیمتیں استعمال کرتا ہے: 0 اور 1، جنہیں بیت (Bit) کہتے ہیں۔ 0 کا مطلب بند، کم یا غلط ہے اور 1 کا مطلب چالو، زیادہ یا درست ہے۔ ڈیجیٹل نظام قابلِ پھروسا، درست، بنانے میں آسان اور سافٹ ویئر سے قابو میں رہنے والے ہوتے ہیں۔

CHECK YOURSELF

1. What does a 1 represent in a digital system?  
2. Give two advantages of digital systems.

Answers: 1. ON, HIGH or TRUE. 2. Any two of: reliable (less affected by noise), accurate (no loss when copying), easy to design (two states), programmable (controlled by software).

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. A discrete quantity is:  
a. Counted in separate steps  
b. Any value on a smooth range  
c. Always analog  
d. A sine wave

MCQ 2. Temperature of tea is:  
a. Continuous  
b. Discrete  
c. Only Boolean  
d. A gate

MCQ 3. Number of students in a class is:  
a. Discrete  
b. Continuous  
c. Analog voltage  
d. A ramp

MCQ 4. The stairs vs ramp analogy: stairs are:  
a. Discrete  
b. Continuous  
c. TCP  
d. Agile

MCQ 5. Digital systems use:  
a. 0 and 1  
b. Every real number  
c. Only sine waves  
d. Paper only

MCQ 6. A bit is:  
a. A binary digit, 0 or 1  
b. Eight bytes  
c. A logic gate  
d. A database

Extra short questions

1. Differentiate discrete and continuous with one example each.  
2. Why do computers prefer digital values?  
3. What is a digital system?  
4. Give two advantages of digital systems.

Extra long question

1. With a labelled stairs/ramp sketch, explain discrete vs continuous and link both to analog vs digital signals.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Discrete: countable steps (cars in a park). Continuous: any value in a range (speed of a car).  
2. Bits 0/1 can be restored after noise; analog waves are bent by interference.  
3. An electronic system that stores and processes only two values, 0 and 1.  
4. Reliable (noise resistant), accurate copies, easy to design (two states), programmable.

Long

1. Stairs = discrete/digital (fixed levels). Ramp = continuous/analog (smooth). Computers convert analog measurements into bits, process bits, then convert back if a human must see the result.

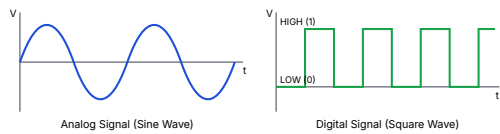
1.1.3 Analog and Digital Signals ★ Golden

LEARN IT

A **signal** is a changing voltage (or other quantity) that carries information.

**Analog signals** are continuous signals that change smoothly over time, like a **sine wave**. Examples: the human voice, a traditional mercury thermometer.

**Digital signals** have only two levels: **HIGH (1)** and **LOW (0)**. They jump instantly between these values, which creates a **square wave**. Example: computer data.



An analog signal changes smoothly; a digital signal jumps between only two levels.

| Feature     | Analog data                           | Digital data                      |
|-------------|---------------------------------------|-----------------------------------|
| Nature      | Continuous flow.                      | Discrete, separate values (bits). |
| Wave shape  | Sine wave (smooth curve).             | Square wave (sudden jumps).       |
| Reliability | Low (affected by noise).              | High (resistant to noise).        |
| Examples    | Human voice, traditional thermometer. | Computer data.                    |

WORKED EXAMPLE

Why is digital more resistant to noise?

Step 1: Suppose a wire carries 5 V for HIGH and 0 V for LOW.

Step 2: Noise adds a small error, so 5 V arrives as 4.6 V and 0 V arrives as 0.3 V.

Step 3: The receiver only asks "Is it closer to HIGH or LOW?" 4.6 V is still read as 1 and 0.3 V is still read as 0. The data is perfect.

Step 4: In an analog signal, every exact value carries meaning, so 4.6 V instead of 5 V really changes the sound or reading. The noise cannot be removed.

REMEMBER IT

Analog = A smooth curve. Digital = Doorsteps. Analog looks like ocean waves; digital looks like a castle wall (square teeth).

COMMON MISTAKE

Do not draw a digital signal with sloping sides. A digital signal jumps *instantly*, so its edges are straight vertical lines. Also do not confuse "digital" with "electronic": an analog signal can also be electronic.

اردو میں سمجھیں

اینا لگ سگنل وقت کے ساتھ ہموار انداز میں بدلتا ہے، جیسے سائن ویو (Sine Wave)؛ انسانی آواز اس کی مثال ہے۔ ڈیجیٹل سگنل کی صرف دو سطحیں ہوتی ہیں، HIGH (1) اور LOW (0)، اور یہ اچانک ایک سے دوسری پر جاتا ہے، اس لیے مربع لہر (Square Wave) بنتی ہے۔ ڈیجیٹل سگنل شور سے کم متاثر ہوتا ہے، اس لیے زیادہ قابلِ بھروسہ ہے۔

EXAM TIP

Common questions: "Differentiate between analog and digital signals" (3–4 marks) and "Draw an analog and a digital signal." Write the comparison as a table with at least three features (nature, wave shape, reliability, example). Always label your diagrams: axes V and t, and HIGH(1)/LOW(0) on the square wave.

CHECK YOURSELF

1. What shape of wave does a digital signal make?

2. Which type of signal is more affected by noise?

3. Give one example of an analog signal.

Answers: 1. A square wave. 2. Analog. 3. Human voice (or a traditional thermometer reading).

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. An analog signal:  
a. Changes smoothly, like a sine wave  
b. Jumps only 0/1  
c. Is a truth table  
d. Is TCP

MCQ 2. A digital signal looks like:  
a. A square wave of HIGH and LOW  
b. A smooth sine  
c. A pie chart  
d. A stack

MCQ 3. Noise harms analog more because:  
a. Many in-between levels can be bent  
b. Bits cannot exist  
c. OSI has 7 layers  
d. Python is analog

MCQ 4. Human voice on a microphone is first:  
a. Analog  
b. A Boolean 1 only  
c. A K-map  
d. SQL

MCQ 5. HIGH in digital means:  
a. 1 / ON / TRUE  
b. 0 / OFF  
c. A continuous ramp  
d. Gray code

MCQ 6. Copying analog tape many times:  
a. Loses quality  
b. Stays perfect  
c. Creates bits  
d. Builds NAND

Extra short questions

1. Define analog signal and digital signal.

2. Give one example of each.

3. Why is digital more reliable?

4. What does a computer do with analog sound?

Extra long question

1. Draw analogue (sine) and digital (square) waves. Tabulate nature and reliability. This is a ★ Golden 10-mark.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Analog: continuous, smooth change. Digital: only two levels, HIGH(1) and LOW(0), a square wave.

2. Analog: traditional thermometer / voice. Digital: computer data / logic pulse.

3. A bent 0 or 1 can be restored to a clean 0 or 1; analog distortion stays in the wave.

4. Sample and quantise it into bits, process the bits, then convert back to analog for a speaker.

Long

1. Sine labelled V against t; square labelled HIGH/LOW. Table: analog = continuous, low reliability; digital = discrete bits, high reliability. Close with the noise argument.

1.1.4 – 1.1.6 Boolean Algebra, Operations & Truth Tables

| LEARN IT                                                                                                                                                                                                                                                                                                                      |                                           |                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|--------------------|
| Boolean algebra was developed by the English mathematician <b>George Boole</b> . It is a kind of algebra that deals with only two values: <b>TRUE</b> (1) and <b>FALSE</b> (0). It is the foundation of digital electronics and computer programming — every 1 of condition in a program is a small piece of Boolean algebra. |                                           |                    |
| Concept                                                                                                                                                                                                                                                                                                                       | Description                               | Example            |
| Variables                                                                                                                                                                                                                                                                                                                     | Letters that hold the value 0 or 1.       | A, B, C, X, Y      |
| Operations                                                                                                                                                                                                                                                                                                                    | Actions performed on variables.           | AND, OR, NOT       |
| Expressions                                                                                                                                                                                                                                                                                                                   | Combinations of variables and operations. | A AND B, Y = A + B |

Some simple rules (laws) follow directly from the meaning of AND, OR and NOT. They are useful when you check or simplify answers:

| OR rules    | AND rules       | NOT rules                       |
|-------------|-----------------|---------------------------------|
| $A + 0 = A$ | $A \cdot 1 = A$ | $(\overline{\overline{A}}) = A$ |
| $A + 1 = 1$ | $A \cdot 0 = 0$ | $A + \overline{A} = 1$          |
| $A + A = A$ | $A \cdot A = A$ | $A \cdot \overline{A} = 0$      |

WORKED EXAMPLE

Using a law: Simplify  $Y = AB + AB$ .  
Step 1 Both terms contain B, so take B out:  $Y = B(A + A)$ .  
Step 2  $A + A = 1$  (a variable OR its opposite is always 1).  
Step 3  $Y = B \cdot 1 = B$ .

COMMON MISTAKE

In Boolean algebra “+” means OR, not addition. So  $1 + 1 = 1$  (TRUE OR TRUE is TRUE), not 2. There is no value bigger than 1.

آردو میں سمجھیں

جارج بول (George Boole) نے بنایا۔ اس میں صرف دو فیصلے ہوتے ہیں: درست (1) اور غلط (0)۔ متغیرات (Variables) جیسے A اور B صرف 0 یا 1 رکھتے ہیں، آپریٹرز AND، OR اور NOT ہیں اور ان کی مکالمے سے ایکسپریشن (Expressions) بنتا ہے۔

CHECK YOURSELF

1. Who developed Boolean algebra?

2. In Boolean algebra, what is  $1 + 0$ ?

Answers: 1. George Boole. 2. 1 (Because + means OR).

LEARN IT

There are three basic Boolean operations. Every other operation can be made from them.

1. **AND operation** ( $\cdot$ ) — gives output 1 **only when ALL inputs are 1**. Expression:  $\overline{Y} = \overline{A \cdot B}$  (often written just AB).

2. **OR operation** ( $+$ ) — gives output 1 **when ANY input is 1**. Expression:  $Y = A + B$ .

3. **NOT operation** ( $\overline{\phantom{x}}$  or  $\neg$ ) — **inverts** (flips) the input. If the input is 1 the output is 0, and if the input is 0 the output is 1. It is a **unary** operation, meaning it works on only one operand. Expression:  $Y = \overline{A}$  (also written as A with a bar on top,  $\overline{A}$ ).

| Operation | Real life analogy                                                                                                                                               |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AND       | A door with two locks. The door opens only when <b>both</b> locks are unlocked.                                                                                 |
| OR        | A room with two doors. You can enter if <b>either</b> door is open.                                                                                             |
| NOT       | A room light switch. When the switch is <b>ON</b> the room is <b>lit</b> ; when it is <b>OFF</b> the room is <b>dark</b> — the state is the opposite of “dark”. |

WORKED EXAMPLE

Evaluate each expression for  $A = 1$ ,  $B = 0$ .

(a)  $A \cdot B + 1 \cdot 0 = 0$  (not all inputs are 1).

(b)  $A + B + 1 + 0 = 1$  (at least one input is 1).

(c)  $\overline{B} + \overline{A} = 1$  (NOT flips 0 to 1).

(d)  $A \cdot \overline{B} = 1 \cdot 1 = 1$

REMEMBER IT

AND is “strict” — everyone must agree (all 1s). OR is “easy-going” — one 1 is enough. NOT is “the rebel” — it always says the opposite.

آردو میں سمجھیں

AND آپریٹرز میں 1 بننا ہے جب تمام ان پٹ 1 ہیں (دونوں دروازے OR آپریٹرز میں 1 بننا ہے جب کوئی ایک ان پٹ بھی 1 ہو (دونوں دروازے ولاک کردہ) NOT آپریٹرز ان پٹ کو الٹ دیتا ہے اور صرف ایک ان پٹ پر کام کرتا ہے، اس لیے آسان پوچھ کر (Logic) کہتے ہیں۔

CHECK YOURSELF

1. What symbol is used for the OR operation?

2. Why is NOT called a unary operation?

3. Find  $A + \overline{B}$  when  $A = 0$ ,  $B = 1$ .

Answers: 1. +. 2. Because it works on only one operand (one input). 3.  $\overline{B} = 0$ , so  $0 + 0 = 0$ .

LEARN IT

A truth table shows **all possible input combinations** and the output for each one, for a logic gate or a circuit. It is the “complete record” of how the circuit behaves.

How to create a truth table:

- Count the number of inputs (n).
- Calculate total rows =  $2^n$ .
- List all input combinations in binary counting order (0, 1, 2, 3 ... in binary).
- Calculate the output for each combination.

2-input example ( $n = 2$ , rows = 4)

| A | B | Combination |
|---|---|-------------|
| 0 | 0 | Row 1       |
| 0 | 1 | Row 2       |
| 1 | 0 | Row 3       |
| 1 | 1 | Row 4       |

3-input example ( $n = 3$ , rows = 8)

| A | B | C | Decimal |
|---|---|---|---------|
| 0 | 0 | 0 | 0       |
| 0 | 0 | 1 | 1       |
| 0 | 1 | 0 | 2       |
| 0 | 1 | 1 | 3       |
| 1 | 0 | 0 | 4       |
| 1 | 0 | 1 | 5       |
| 1 | 1 | 0 | 6       |
| 1 | 1 | 1 | 7       |

REMEMBER IT

**Quick fill trick for 3 inputs:** In column C write 0,0,1,1... (change every row). In column B write 0,0,1,1,1,1 (change every 2 rows). In column A write 0,0,0,0,1,1,1,1 (change every 4 rows). For n inputs the leftmost column changes every  $2^{n-1}$  rows.

WORKED EXAMPLE

Make the truth table of a 3-input AND  $\overline{Y} = \overline{A \cdot B \cdot C}$ .

Step 1:  $n = 3$ , so rows =  $2^3 = 8$ .

Step 2: List the 8 combinations in binary order.

Step 3: AND gives 1 only when all inputs are 1, so only row 7 (111) gives 1.

| A | B | C | $Y = A \cdot B \cdot C$ |
|---|---|---|-------------------------|
| 0 | 0 | 0 | 0                       |
| 0 | 0 | 1 | 0                       |
| 0 | 1 | 0 | 0                       |
| 0 | 1 | 1 | 0                       |
| 1 | 0 | 0 | 0                       |
| 1 | 0 | 1 | 0                       |
| 1 | 1 | 0 | 0                       |
| 1 | 1 | 1 | 1                       |

COMMON MISTAKE

Do not fill rows in a random order — you may miss or repeat a combination. Always count in binary from 000 to 111. Also remember rows =  $2^n$ , not  $2 + n$ . 3 inputs give 8 rows, not 6.

آردو میں سمجھیں

درجہ قبل (Truth Table) کسی گیت یا سرکٹ کی تمام ممکنہ ان پٹ اور ان کی آؤٹ پٹ دکھاتا ہے۔ اگر ان پٹ 4 ہیں تو قطاریں 2 (Rows) کی طاقت 4 ہیں گی، دو ان پٹ کی آئی 4 اور تین ان پٹ کی آئی 8 قطاریں، ان پٹ بیکار یا غلطی نہیں نکلیں۔

CHECK YOURSELF

1. How many rows does a truth table with 4 inputs have?

2. In a 3-input table, what is the 8th row (counting from row 1 = 000)?

Answers: 1.  $2^4 = 16$ . 2. 101.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

|                                                                                                                         |                                                                                                                           |                                                                                                                           |                                                            |                                                                                           |                                                                                               |
|-------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|-------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| <b>MCQ 1.</b> Boolean algebra uses:<br>a. TRUE/F and FALSE/0 only<br>b. All real numbers<br>c. ASCII letters<br>d. IPv4 | <b>MCQ 2.</b> AND is 1 only when:<br>a. Every input is 1<br>b. Any input is 1<br>c. Inputs differ<br>d. There is no input | <b>MCQ 3.</b> OR is 1 when:<br>a. Any input is 1<br>b. All inputs are 0<br>c. Inputs are equal<br>d. NOT is applied twice | <b>MCQ 4.</b> NOT of 1 is:<br>a. 0<br>b. 1<br>c. 2<br>d. A | <b>MCQ 5.</b> Rows in an n-input truth table:<br>a. $2^n$<br>b. n<br>c. $n^2$<br>d. $n+1$ | <b>MCQ 6.</b> Two-input truth table has:<br>a. 4 rows<br>b. 2 rows<br>c. 8 rows<br>d. 16 rows |
|-------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|-------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|

Extra short questions

- Who developed Boolean algebra and what two values does it use?
- With  $\overline{Y} = \overline{A \cdot B}$  and  $Y = A + B$  in words.
- Why is NOT called unary?
- List the four steps to build a truth table.

Extra long question

- Write complete truth tables for 2-input AND, OR and 1-input NOT. Add one real-life lock/door/switch example each.

Extra answer key

**MCQs:** 1. A, 2. A, 3. A, 4. A, 5. A, 6. A

Short

- George Boole, TRUE/F and FALSE/0.
- AND: output 1 only if all inputs are 1. OR: output 1 if any input is 1.
- It has only one operand: it inverts a single input.
- Count inputs n; rows =  $2^n$ ; list combinations in binary order; compute each output.

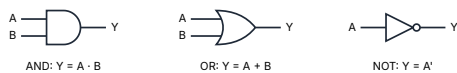
Long

- AND: 0001 on AB=11 (two locks), OR: 0111 (two doors), NOT: 10 / 01 (light switch). Show all four-two rows.

## 1.1.7 Basic Logic Gates (AND, OR, NOT) ★ Golden

### LEARN IT

**Logic gates** are electronic circuits that perform Boolean operations. They take one or more input bits and produce one output bit. They are the **building blocks of all digital systems** — a processor contains billions of them. The three basic gates match the three basic operations: the **AND gate**, the **OR gate** and the **NOT gate** (also called an **inverter**).



Symbols of the basic gates. AND has a flat back and round front ("D" shape); OR has a curved back and pointed front; NOT is a triangle with a small circle (bubble).

### AND gate — $Y = A \cdot B$

| A | B | Y |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

### OR gate — $Y = A + B$

| A | B | Y |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 1 |

### NOT gate — $Y = A'$

| A | Y |
|---|---|
| 0 | 1 |
| 1 | 0 |

### WORKED EXAMPLE

**A school bell circuit.** The bell (Y) should ring when the timer says "period over" ( $A = 1$ ) **and** the school is open ( $B = 1$ ).

Step 1: The bell needs both conditions, so use an AND gate:  $Y = A \cdot B$ .

Step 2: Sunday, timer fires:  $A = 1$ ,  $B = 0 \rightarrow Y = 1 \cdot 0 = 0$ . No bell.

Step 3: Monday, timer fires:  $A = 1$ ,  $B = 1 \rightarrow Y = 1$ . The bell rings.

If instead the bell should ring when *either* the timer fires or the principal presses a button, you would use an OR gate.

### REMEMBER IT

**Shortcut:** AND output is 0 if any input is 0. OR output is 1 if any input is 1. Learn just the "special row": AND is 1 only at 11; OR is 0 only at 00.

### COMMON MISTAKE

Do not mix up the AND and OR symbols. AND has a **flat** input side (like the letter D). OR has a **curved** input side and a **pointed** output, like a shield or arrowhead.

اردو میں سمجھیں

لاچک گیٹس الیکٹرانک سرکٹ میں جو پولین آپریشن انجام دیتے ہیں اور تمام ڈیجیٹل نظاموں کی بنیادی اینٹیں ہیں۔ AND گیٹ صرف تب 1 دیتا ہے جب دونوں ان پٹ 1 ہوں، OR گیٹ صرف تب 0 دیتا ہے جب دونوں ان پٹ 0 ہوں، اور NOT گیٹ ان پٹ کو الٹ دیتا ہے۔

### EXAM TIP

You are often asked to "draw the symbol, write the expression and truth table" of a gate (2–3 marks each). Draw neat symbols with labelled inputs A, B and output Y. Write the full truth table in binary order. A frequent MCQ: "Which gate gives 1 only when all inputs are 1?" (AND).

### CHECK YOURSELF

- What is the output of an OR gate when  $A = 0$  and  $B = 0$ ?
- What is another name for the NOT gate?
- For which input does a 2-input AND gate output 1?

Answers: 1. 0. 2. Inverter. 3. Only  $A = 1$ ,  $B = 1$ .

### Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

#### MCQ 1. Logic gates implement:

- Boolean operations in hardware
- SQL joins
- HTTP only
- Pie charts

#### MCQ 2. AND output is 1 for inputs 1,1:

- 1
- 0
- Z
- X

#### MCQ 3. OR of 1 and 0 is:

- 1
- 0
- NAND
- XOR

#### MCQ 4. The inverter is the:

- NOT gate
- AND gate
- Bus
- Router

#### MCQ 5. Building blocks of digital systems are:

- Logic gates
- Spreadsheets
- MVPs
- Fonts

#### MCQ 6. $Y = A \cdot B$ is the:

- AND expression
- OR expression
- XOR
- NOR

### Extra short questions

- Define a logic gate.
- State the AND, OR, NOT expressions.
- Draw the truth table of 2-input AND from memory.
- Why is NOT drawn with a bubble?

### Extra long question

- For AND, OR and NOT: symbol, expression, full truth table, and one daily-life model. ★ Golden.

#### Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

#### Short

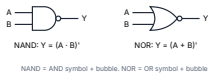
- An electronic circuit that performs a Boolean operation; the building block of digital systems.
- $Y = A \cdot B$ ;  $Y = A + B$ ;  $Y = A'$  (or  $\bar{A}$ ).
- 00→0, 01→0, 10→0, 11→1.
- The bubble marks inversion of the input.

#### Long

- Standard IEC/US symbols. AND two locks, OR two doors, NOT a switch. Tables must have every row.

1.1.7 Universal & Advanced Logic Gates

**LEARN IT**  
**NAND** means "NOT AND": it is an AND gate followed by a **NOT**. **NOR** means "NOT OR": an OR gate followed by a **NOT**. On the symbol, the NOT is shown as a small bubble at the output.  
NAND and NOR are called **universal gates** because **any logic circuit in the world can be built using only NAND gates (or only NOR gates)**. Factories like this: they can mass-produce one type of chip and build everything from it.



**NAND (NOT AND) —  $Y = (A \cdot B)'$**

| A | B | AB | Y |
|---|---|----|---|
| 0 | 0 | 0  | 1 |
| 0 | 1 | 0  | 1 |
| 1 | 0 | 0  | 1 |
| 1 | 1 | 1  | 0 |

**NOR (NOT OR) —  $Y = (A + B)'$**

| A | B | A+B | Y |
|---|---|-----|---|
| 0 | 0 | 0   | 1 |
| 0 | 1 | 1   | 0 |
| 1 | 0 | 1   | 0 |
| 1 | 1 | 1   | 0 |

**WORKED EXAMPLE**  
Show that a **NAND gate can work as a NOT gate**.  
Step 1: Connect the same input A to both inputs of the NAND gate. Then  $Y = (A \cdot A)'$ .  
Step 2: Since  $A \cdot A = A$ , we get  $Y = A'$ .  
Step 3: Check:  $A = 0 \rightarrow (0 \cdot 0)' = 0' = 1$ ,  $A = 1 \rightarrow (1 \cdot 1)' = 1' = 0$ . This is exactly the NOT truth table.  
In the same way, a NOR gate with both inputs tied together gives  $(A + A)' = A'$ . And a NAND followed by a NAND-as-NOT gives  $((A \cdot B)')' = A \cdot B$ , an AND gate. This is why they are "universal".

**REMEMBER IT**  
NAND is the exact **opposite** of AND (0 only at 11). NOR is the exact opposite of OR (1 only at 00). Just flip the AND/OR output column.

**COMMON MISTAKE**  
Do not forget the bubble. Without the small circle, a NAND symbol becomes an AND symbol and your answer is wrong. Also,  $(A \cdot B)'$  is **not** the same as  $A' \cdot B'$ : for  $A = 1, B = 0$ ,  $(1 \cdot 0)' = 1$  but  $0 \cdot 1 = 0$ .

آردو میں سمجھیں

NAND کا مطلب ہے NOT AND اور NOR کا مطلب ہے NOT OR. انہیں یونیورسل (عامگیر) گیت کہتے ہیں کیونکہ صرف NAND یا صرف NOR گیتوں سے دنیا کا کوئی بھی لایک سرکٹ بنایا جا سکتا ہے۔ NAND صرف تب 0 دیتا ہے جب دونوں ان پٹ 1 ہوں، اور NOR صرف تب 1 دیتا ہے جب دونوں ان پٹ 0 ہوں۔

**CHECK YOURSELF**  
1. Why are NAND and NOR called universal gates?  
2. What is the NOR output for  $A = 0, B = 1$ ?  
3. How do you make a NOT gate from a NAND gate?  
Answers: 1. Any logic circuit can be built using only NAND gates or only NOR gates. 2. 0. 3. Join both inputs together and feed A into them:  $(A \cdot A)' = A'$ .

**LEARN IT**  
**XOR (Exclusive OR)** gives output 1 when the inputs are **different**. Expression:  $Y = A \oplus B$ . In terms of basic operations,  $A \oplus B = A'B + AB'$ .  
**XNOR (Exclusive NOR)** is XOR followed by NOT. It gives output 1 when the inputs are the **same**. Expression:  $Y = (A \oplus B)' = A'B' + AB$ .  
The XOR symbol is an OR symbol with an extra curved line at the input side. XNOR adds a bubble at the output.



**XOR —  $Y = A \oplus B$**

| A | B | Y |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

**XNOR —  $Y = (A \oplus B)'$**

| A | B | Y |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

**All seven gates at a glance**

| A | B | AND | OR | NAND | NOR | XOR | XNOR |
|---|---|-----|----|------|-----|-----|------|
| 0 | 0 | 0   | 0  | 1    | 1   | 0   | 1    |
| 0 | 1 | 0   | 1  | 1    | 0   | 1   | 0    |
| 1 | 0 | 0   | 1  | 1    | 0   | 1   | 0    |
| 1 | 1 | 1   | 1  | 0    | 0   | 0   | 1    |

NOT has one input:  $A \cdot 0 = 1, A \cdot 1 = 0$

**WORKED EXAMPLE**  
**A staircase light.** A light has one switch at the bottom (A) and one at the top (B) of a staircase. Flipping either switch should change the light.  
Step 1: Start with both switches down:  $A = 0, B = 0$ , light off ( $Y = 0$ ).  
Step 2: Flip the bottom switch:  $A = 1, B = 0 \rightarrow$  light on ( $Y = 1$ ).  
Step 3: Walk up and flip the top switch:  $A = 1, B = 1 \rightarrow$  light off ( $Y = 0$ ).  
Step 4: The pattern 0, 1, 1, 0 is the XOR truth table. So the light is  $Y = A \oplus B$ .

**REMEMBER IT**  
Memory trick: XOR outputs 1 when inputs are "exclusively" different. XNOR outputs 1 when inputs are the **same**. XNOR is sometimes called the "equality gate".

**COMMON MISTAKE**  
XOR is not the same as OR. They differ only in the last row: for  $A = 1, B = 1$ , OR gives 1 but XOR gives 0.

آردو میں سمجھیں

XOR گیت تب 1 دیتا ہے جب دونوں ان پٹ مختلف ہوں، اور XNOR گیت تب 1 دیتا ہے جب دونوں ان پٹ ایک جیسے ہوں۔ XNOR براہِ اصل XOR کے بعد NOT ہے۔ OR اور XOR میں فرق صرف آخری قطار میں ہے جہاں دونوں ان پٹ 1 ہوں۔

**CHECK YOURSELF**  
1. What is  $1 \oplus 1$ ?  
2. Which gate outputs 1 when both inputs are the same?  
3. Write XOR using AND, OR and NOT.  
Answers: 1. 0. 2. XNOR. 3.  $A \oplus B = A'B + AB'$ .

**Extra board practice**  
These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

**MCQ 1.** NAND and NOR are universal because:

- Any circuit can be built from only that gate
- They are analog
- They store files
- They sort arrays

**MCQ 2.** NAND is:

- NOT of AND
- AND of NOT
- OR of XOR
- Only OR

**MCQ 3.** NAND of 1,1 is:

- 0
- 1
- Z
- 2

**MCQ 4.** NOR of 0,0 is:

- 1
- 0
- A
- B

**MCQ 5.** XOR is 1 when inputs are:

- Different
- Both 1 only
- Both 0 only
- Always 1

**MCQ 6.** XNOR is 1 when inputs are:

- The same
- Different
- Only 1,0
- Flipping

**Extra short questions**  
1. Why are NAND and NOR called universal gates?  
2. Write NAND and NOR expressions.  
3. Memory trick for XOR vs XNOR.  
4. Truth table of XOR.

**Extra long question**  
1. Show how a NOT, AND and OR can each be built from NAND only. Include truth-table checks.

**Extra answer key**  
MCQs: 1. A 2. A 3. A 4. A 5. A 6. A  
**Short**  
1. Any Boolean function (AND, OR, NOT, XOR...) can be built using only NAND, or only NOR.  
2.  $Y = (A \cdot B)'$  and  $Y = (A + B)'$ .  
3. XOR: exclusively different  $\rightarrow$  1. XNOR: same  $\rightarrow$  1 (equivalence).  
4. 00 $\rightarrow$ 0, 01 $\rightarrow$ 1, 10 $\rightarrow$ 1, 11 $\rightarrow$ 0.  
**Long**  
1. NOT: NAND with inputs tied. AND: NAND then NAND-NOT. OR: invert inputs with NAND-NOTs then NAND (De Morgan). Tables must match the target gate.

11.8 – 11.11 Logic Expressions, Minterms & Diagrams

**LEARN IT**

A **logic expression** (Boolean expression) combines variables with operations, for example  $Y = A + B \cdot C$ . To evaluate it correctly you must follow the **order of precedence**, just the BODMAS in maths.

Parenttheses ( )

2. NOT ( )

3. AND ( )

4. OR ( + )

**WORKED EXAMPLE**

**Worked example 1:** truth table of  $Y = A + B \cdot C$

Step 1: Inputs, six rows.

Step 2: Precedence: first find  $B \cdot C$  (NOT), then  $B \cdot C$  (AND), then  $A + B \cdot C$  (OR). Make a column for each step.

| A | B | C | B · C | Y = A + B · C |
|---|---|---|-------|---------------|
| 0 | 0 | 0 | 0     | 0             |
| 0 | 0 | 1 | 0     | 0             |
| 0 | 1 | 0 | 0     | 0             |
| 0 | 1 | 1 | 1     | 1             |
| 1 | 0 | 0 | 0     | 1             |
| 1 | 0 | 1 | 0     | 1             |
| 1 | 1 | 0 | 0     | 1             |
| 1 | 1 | 1 | 1     | 1             |

Step 3:  $Y = 1$  in rows 1, 4, 5, 6, 7 (decimal). Check one row by hand:  $A=0$ ,  $B=0$ ,  $C=1$ ,  $B \cdot C = 0$ ,  $Y = 0 + 0 = 0$ .

**WORKED EXAMPLE**

**Worked example 2:** truth table of  $Y = (A + B) \cdot C$

Step 1: Parentheses first: find  $A + B$ . Then NOT  $C$ . Then AND the two.

| A | B | C | A + B | C' | Y |
|---|---|---|-------|----|---|
| 0 | 0 | 0 | 0     | 1  | 0 |
| 0 | 0 | 1 | 0     | 0  | 0 |
| 0 | 1 | 0 | 1     | 1  | 0 |
| 0 | 1 | 1 | 1     | 0  | 0 |
| 1 | 0 | 0 | 1     | 1  | 0 |
| 1 | 0 | 1 | 1     | 0  | 0 |
| 1 | 1 | 0 | 1     | 1  | 0 |
| 1 | 1 | 1 | 1     | 0  | 0 |

Step 2:  $Y = 1$  in rows 2, 4 and 6 (decimal).

**LEARN IT**

Sometimes you are given a truth table and must find the expression. There are two standard ways.

**Minterms** based on SOP – Sum of Products – product (AND) of all the variables, written for a row whose output is 1. A variable is complemented (A') if its value is 0, and uncomplemented (A) if its value is 1. OR-ing all the minterms gives the SOP expression.

**Maxterms** based on POS – Product of Sums – sum (OR) of all the variables, written for a row whose output is 0. A variable is complemented (A') if its value is 1, and uncomplemented (A) if its value is 0. AND-ing all the maxterms gives the POS expression.

| A | B | C | Output (Y) | Minterm (SOP) | Maxterm (POS) |
|---|---|---|------------|---------------|---------------|
| 0 | 0 | 0 | 0          | –             | $A + B$       |
| 0 | 0 | 1 | 1          | $AB$          | –             |
| 0 | 1 | 0 | 0          | –             | $A + B$       |
| 0 | 1 | 1 | 1          | $AB$          | –             |
| 1 | 0 | 0 | 1          | $AB$          | –             |
| 1 | 0 | 1 | 1          | $AB$          | –             |
| 1 | 1 | 0 | 1          | $AB$          | –             |
| 1 | 1 | 1 | 1          | $AB$          | –             |

Your SOP expression:  $Y = AB + AB + AB + AB + AB + AB + AB + AB$  (that POS expression:  $Y = (A + B)(A + B)$  both describe the same table. (Both simplify to  $Y = B$  – look at the table: Y always equals B. You will confirm this with a K-map later).

**WORKED EXAMPLE**

**Worked example 3:** write the SOP for a 3-input "majority" circuit.  $Y = 1$  when at least two of A, B, C are 1.

| Row | A | B | C | Y | Minterm |
|-----|---|---|---|---|---------|
| 0   | 0 | 0 | 0 | 0 | –       |
| 1   | 0 | 0 | 1 | 0 | –       |
| 2   | 0 | 1 | 0 | 0 | –       |
| 3   | 0 | 1 | 1 | 1 | ABC     |
| 4   | 1 | 0 | 0 | 0 | –       |
| 5   | 1 | 0 | 1 | 1 | ABC     |
| 6   | 1 | 1 | 0 | 1 | ABC     |
| 7   | 1 | 1 | 1 | 1 | ABC     |

Step 1: Pick the rows with  $Y = 1$ : rows 3, 5, 6, 7.

Step 2: Row 3 is  $A=0$ ,  $B=1$ ,  $C=1$  so  $A$  is 0 so write  $A'$ ; B and C are 1 so write B, C:  $ABC$ .

Step 3: Do the same for the other rows and OR them:  $Y = ABC + ABC + ABC + ABC$  (in short form:  $Y = \Sigma m(3,5,6,7)$ ) this will simplify this with a K-map to 1111.

$Y = ABC + ABC + ABC + ABC = \Sigma m(3,5,6,7)$

**REMEMBER IT**

SOP → Sum of Products. OR, Minterms: 0 → bad, 1 → good. Maxterms: the reverse (1 → bad, 0 → good). Think "Min: Minus, Max: Plus".

**COMMON MISTAKES**

The classic errors (1) writing minterms for the 0-rows – minterms come only from rows where  $Y = 1$  (2) forgetting precedence →  $AB + B \cdot C$  is not  $(A + B) \cdot C$  for  $A=1$ ,  $B=0$ ,  $C=0$  the first gives 1 and the second gives 0.

از این متن استفاده کنید

در این متن، ما به شما یاد می‌دهیم که چگونه یک عبارت منطقی را به یک جدول حقیقت تبدیل کنید. این فرآیند شامل شناسایی متغیرها، تعیین عملیات منطقی و استفاده از اولویت عملیات است. ما همچنین به شما یاد می‌دهیم که چگونه یک جدول حقیقت را به یک عبارت منطقی تبدیل کنید. این فرآیند شامل شناسایی سطرها که خروجی 1 دارند و نوشتن عبارت منطقی برای هر یک از این سطرها است. در نهایت، ما به شما یاد می‌دهیم که چگونه یک عبارت منطقی را به یک مدار منطقی تبدیل کنید. این فرآیند شامل شناسایی عملیات منطقی و استفاده از نمادهای استاندارد است.

**CHECK YOURSELF**

1. Which comes first: AND or OR?

2. Write the minterms for  $A + B \cdot C$  and  $A \cdot B + C$ .

3. Write the maxterms for  $A + B \cdot C$  and  $A \cdot B + C$ .

4. Write the minterms for  $A + B \cdot C$  and  $A \cdot B + C$ .

**LEARN IT**

A **logic diagram** shows how logic gates are connected to create a circuit. It is a "picture" of a Boolean expression. To draw one, use the order of precedence: the operation with the **highest priority** is drawn first (nearest the inputs), and the last operation is the gate that gives the final output.

**WORKED EXAMPLE**

**Example 1:** Draw the logic diagram for  $Y = A \cdot B + C$ .

Step 1: Find the operations: AND (·) and OR (+).

Step 2: AND has higher precedence than OR, so draw the AND gate first, with inputs A and B. Its output is  $A \cdot B$ .

Step 3: Draw an OR gate. Feed  $A \cdot B$  and C into the AND gate to get Y.

$Y = A \cdot B + C$  (OR)  $A \cdot B$  (AND) first, then OR.

**WORKED EXAMPLE**

**Example 2:** Draw the logic diagram for  $Y = (A + B) \cdot C$ .

Step 1: Parentheses have the highest priority, so draw the OR gate for  $A + B$  first.

Step 2: NOT is next: draw a NOT gate on input C to get  $C'$ .

Step 3: The last operation is AND: feed  $(A + B)$  and  $C'$  into an AND gate to get Y.

$Y = (A + B) \cdot C'$  (OR and NOT first, then AND).

You can check this circuit against the truth table in 11.8 (worked example 2):  $Y = 1$  only for rows 010, 100 and 110.

**REMEMBER IT**

Read the expression and ask "What is done last?" That gate goes on the far right, next to Y. Work backwards to the inputs. Count gates:  $Y = AB + C$  needs 2 gates;  $Y = (A + B) \cdot C$  needs 3.

**COMMON MISTAKES**

Do not draw a separate wire for a variable each time it appears and then leave one unconnected. Every gate input must be joined to an input variable or to another gate's output, and every wire should be clearly labelled as inputs.

از این متن استفاده کنید

در این متن، ما به شما یاد می‌دهیم که چگونه یک عبارت منطقی را به یک مدار منطقی تبدیل کنید. این فرآیند شامل شناسایی عملیات منطقی و استفاده از نمادهای استاندارد است. ما همچنین به شما یاد می‌دهیم که چگونه یک مدار منطقی را به یک عبارت منطقی تبدیل کنید. این فرآیند شامل شناسایی عملیات منطقی و استفاده از نمادهای استاندارد است.

**CHECK YOURSELF**

1. In  $Y = A + B \cdot C$ , which gate is drawn first?

2. How many gates are needed for  $Y = A \cdot B + C$ ?

3. How many gates are needed for  $Y = (A + B) \cdot C$ ?

4. How many gates are needed for  $Y = (A + B) \cdot C$ ?

**EXTRA BOARD PRACTICE**

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

**MCQ 1:** Precedence (highest first) is:

1. Parentheses, NOT, AND, OR
2. OR, AND, NOT, AND, OR
3. AND, NOT, OR, AND, OR
4. NOT, AND, OR, AND, OR

**MCQ 2:** A minterm is used when Y is:

1. Always 1
2. Always 0
3. Always 1
4. Always 0

**MCQ 3:** For  $AB + C$  if  $Y = 1$  the minterms are:

1.  $AB$
2.  $AB$
3.  $AB$
4.  $AB$

**MCQ 4:** A minterm is a:

1. Sum of products
2. Product of sums
3. Sum of minterms
4. Product of maxterms

**MCQ 5:** SOP means:

1. Sum of products
2. Sum of minterms
3. Sum of maxterms
4. Sum of products

**MCQ 6:** In a logic diagram you draw first the operation with:

1. Highest precedence
2. Lowest precedence
3. Most inputs
4. Green ink

**Extra short questions**

1. State operator precedence in Boolean algebra.
2. Define minterms and maxterms.
3. From Y rows 01 and 11, write SOP.
4. For  $Y = A + B \cdot C$ , which gate is drawn first?

**Extra long question**

1. Given a 2-input truth table with  $Y = 0,0,1,1$ , write minterms, SOP, maxterms that apply, and draw the logic diagram.

**Extra answer key**

MCQs 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100.

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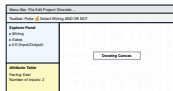


LogiSim Simulation Tool Guide

LEARN IT

LogiSim Evolution is a free graphical tool for designing and simulating logic circuits. "Simulating" means the computer pretends to be the real circuit: you can switch inputs on and off and watch the output change. It lets you see the truth tables and Boolean expressions you learn in theory.

1. The Interface areas



The main areas of the LogiSim Evolution window.

| Area                          | What it does                                                                           |
|-------------------------------|----------------------------------------------------------------------------------------|
| Menu Bar (top)                | File, Edit, Project, Simulate menus — open/save files and control the simulation.      |
| Toolbar                       | Quick access to standard tools: Wiring, Poke, and AND, OR, NOT gates.                  |
| Explorer Panel (left)         | Contains the component libraries: Gates, I/O, Wiring.                                  |
| Attribute Table (bottom left) | Adjust the properties of the selected gate, e.g. Number of inputs or Facing direction. |
| Drawing Canvas (right)        | The dotted grid where you place and wire the gates.                                    |

Wiring colours during simulation

| Colour       | Meaning                   |
|--------------|---------------------------|
| Bright green | Logic 1 (HIGH)            |
| Dark green   | Logic 0 (LOW)             |
| Blue         | Unknown / unconnected pin |
| Red          | Error / short circuit     |

REMEMBER IT

Colours like a traffic light: Green means "working" (bright = 1, dark = 0). Blue means "the wire is lonely — not connected". Red means "stop, something is wrong".

COMMON MISTAKE

If your output wire is blue, the circuit is not wrong in logic — a wire is simply not touching a pin. Zoom in and make sure every wire end sits exactly on a connection point.

اردو میں سمجھیں

لاچی بیم ایپولیفون (LogiSim Evolution) لاجیک سرکٹ بنانی اور چلا کر دیکھنے کا سافٹ ویئر ہے۔ اس میں میٹو ہاں ٹول ہاں ایکسیلوور ہیل، اینڈری بیوت ٹیل اور ڈرائنگ کیبوس بوتی ہیں۔ چھکدار سیر تار 1، گہری سیر 0، نیلی تار غیر منسلک اور سرخ تار غلطی کو ظاہر کرتی ہے۔

CHECK YOURSELF

1. Where do you change the number of inputs of a gate?

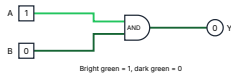
2. What does a red wire mean?

Answers: 1. In the Attribute Table (bottom left). 2. An error or short circuit.

WORKED EXAMPLE

Step by step: building  $Y = A \cdot B$

1. Select the **AND gate** from the Toolbar (or from the Gates library) and place it on the canvas.
2. Select the **Input Pin** (square icon with a 0) and place two pins to the left of the gate. Label them A and B.
3. Select the **Output Pin** (circle icon with a 0) and place it to the right. Label it Y.
4. Use the **Wiring Tool** to click and drag wires from the input pins to the AND gate's input ports. Wires must be horizontal or vertical.
5. Wire the gate's output to the output pin Y.
6. Select the **Poke Tool** (pointing finger) and click inputs A and B to toggle them between 0 and 1. Test the circuit!



Bright green = 1, dark green = 0

Testing with the Poke Tool: A = 1 (bright green wire), B = 0 (dark green), so  $Y = 1 \cdot 0 = 0$ .

WORKED EXAMPLE

Testing the circuit = making its truth table. Poke the inputs through all four combinations and write what Y shows:

| Poke A | Poke B | Y shows | Output wire colour |
|--------|--------|---------|--------------------|
| 0      | 0      | 0       | Dark green         |
| 0      | 1      | 0       | Dark green         |
| 1      | 0      | 0       | Dark green         |
| 1      | 1      | 1       | Bright green       |

This matches the AND truth table, so the circuit is correct. Try the same with  $Y = A \cdot B + C$  from 1.1.1 (add a second gate and a third input pin).

REMEMBER IT

Input pin = square (you push it). Output pin = circle (it shows a light). Poke = finger — you "poke" inputs to change them.

COMMON MISTAKE

Students often try to poke inputs while the Wiring or Select tool is active — then clicking moves the pin instead of changing it. Switch to the Poke Tool first.

اردو میں سمجھیں

لاچی بیم میں  $A \cdot B = Y$  بنانے کے لیے AND گیٹ رکھیں، ہالیں جاب دو ان پٹ ہیں (جوگور) اور دائیں جانب ایک آؤٹ پٹ ہیں (گول) رکھیں، پھر وائرنگ ٹول سے تاریں جوڑیں۔ آخر میں پوک ٹول (انگری) سے ان پٹ کو 0 اور 1 کرکے سرکٹ کو جانچیں۔

CHECK YOURSELF

1. Which icon is the input pin: square or circle?

2. Which tool do you use to change an input between 0 and 1?

Answers: 1. Square (output pin is the circle). 2. The Poke Tool (pointing finger).

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

|                                                                                                                                               |                                                                                                                              |                                                                                                                                |                                                                                                                 |                                                                                                                                       |                                                                                                                                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <b>MCQ 1.</b> LogiSim Evolution is used to:<br>a. Draw and simulate logic circuits<br>b. Edit videos<br>c. Send email<br>d. Compile Java only | <b>MCQ 2.</b> The canvas is where you:<br>a. Place and wire gates<br>b. Install Python<br>c. Write SQL<br>d. Draw pie charts | <b>MCQ 3.</b> Poke tool is for:<br>a. Toggling inputs while simulating<br>b. Deleting the OS<br>c. Printing Urdu<br>d. Sorting | <b>MCQ 4.</b> Attribute table sets:<br>a. Inputs, facing, labels<br>b. Big O<br>c. IP addresses<br>d. Passwords | <b>MCQ 5.</b> To build $Y=A \cdot B$ you need:<br>a. AND gate plus two inputs and an output<br>b. Only XOR<br>c. A stack<br>d. Pandas | <b>MCQ 6.</b> Simulation lets you check:<br>a. The truth table live<br>b. HTTP status<br>c. File modes<br>d. HCI wireframes only |
|-----------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|

Extra short questions

1. Name four areas of the LogiSim window.
2. Steps to build  $Y=A \cdot B$ .
3. Where do you change the number of inputs on a gate?
4. Why simulate at all if the truth table is already on paper?

Extra long question

1. Describe, with a canvas sketch, how you would build and test  $Y = A \cdot B + C$  in LogiSim.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Menu bar, toolbar, explorer (libraries), attribute table, drawing canvas.
2. Place AND; add two input pins; add output pin; wire; label; simulate with Poke.
3. Attribute table (bottom left).
4. Catches wiring mistakes and shows students the table "come alive".

Long

1. AND for A,B; OR with C; three inputs, one output; wiring; poke all 8 combinations; tick the truth table. Mention facing and labels.

1.2.1 – 1.2.3 Software Development Life Cycle (SDLC) ★ Golden

LEARN IT

The **Software Development Life Cycle (SDLC)** is a structured process that guides software development from beginning to end. It ensures **clear goals, better planning, risk reduction and high quality**. Without it, teams forget features, go over budget and deliver buggy software. SDLC has **6 phases**:



Phases 1-3 in detail

| Phase                   | Activities and details                                                                                                                        | Deliverable (what it produces)                     |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| 1. Requirement Analysis | Meet clients, understand their needs, list the features. This is the <b>most important</b> phase — mistakes here spread to every later phase. | SRS Document (Software Requirements Specification) |
| 2. System Design        | Create the High-Level Design (overall architecture), the Low-Level Design (database, UI screens) and flowcharts.                              | Design Document                                    |
| 3. Implementation       | Write the actual code in programming languages (Python, Java, etc.) based on the design documents.                                            | Working Source Code                                |

Phases 4-6 in detail

| Phase          | Activities and details                                               | Deliverable                     |
|----------------|----------------------------------------------------------------------|---------------------------------|
| 4. Testing     | Find and fix bugs (errors), using Black Box and White Box testing.   | Test Reports, bug-free software |
| 5. Deployment  | Release the software to end-users: Direct, Phased, or Pilot release. | Live Software                   |
| 6. Maintenance | Provide support and updates, and fix post-release issues.            | Patches, Updated Versions       |

**Deployment methods:** In a **direct** release the old system is switched off and the new one switched on at once. In a **phased** release the new system is introduced part by part. In a **pilot** release one small group (for example one branch or one class) uses it first before everyone else.

Testing methods

Black Box Testing

- The tester does **not** see the code.
- Tests functionality: give inputs, check outputs.
- Done by dedicated testers.

White Box Testing

- The tester **can** see the code.
- Tests the internal logic and structure.
- Done by developers.

WORKED EXAMPLE

Building a house = SDLC.

- Phase 1 (Requirements): you ask the family what rooms they need — 3 bedrooms, a kitchen, a garage.
- Phase 2 (Design): the architect draws the blueprints.
- Phase 3 (Implementation): the builders lay the bricks.
- Phase 4 (Testing): an inspector checks the wiring and the plumbing.
- Phase 5 (Deployment): the family moves in.
- Phase 6 (Maintenance): repairs, painting and a new room added later.

WORKED EXAMPLE

**Black box vs white box on a login screen.** A black-box tester types a correct password and a wrong password and checks that the first is accepted and the second rejected — without looking at the code. A white-box tester (the developer) reads the code and checks that every `if` branch, for example the "account locked after 3 tries" branch, actually runs correctly.

REMEMBER IT

Remember the order with "Real Developers Implement, Test, Deploy, Maintain" (R-D-I-T-D-M). Black box = a closed box (you can't see inside). White box = a glass box (you can see inside).

COMMON MISTAKE

Do not say the SDLC ends at deployment. Maintenance is a full phase and often lasts the longest. Also, the SRS is the deliverable of Requirement Analysis, not of Design.

آرڈو میں سمجھیں

ایس ڈی ایل سی (SDLC) سافٹ ویئر بنانے کا منظم طریقہ ہے جس کے چھ مراحل ہیں: ضروریات کا تجزیہ، سسٹم ڈیزائن، کوڈنگ (Implementation)، ٹیسٹنگ، تنصیب (Deployment) اور دیکھ بھال (Maintenance)۔ پہلا مرحلہ سب سے اہم ہے اور اس سے SRS دستاویز بنتی ہے۔ ہلکے ہلکے ٹیسٹنگ میں کوڈ نظر نہیں آتا جبکہ وائٹ باکس ٹیسٹنگ میں کوڈ دیکھ کر اندرونی منطقی جانچی جاتی ہے۔

EXAM TIP

"Explain the phases of SDLC" is a very common long question (5-8 marks). Name all 6 phases in order, write 1-2 lines of activities for each, and state the deliverable of each (SRS, Design Document, Source Code, Test Reports, Live Software, Patches). A short question often asks you to differentiate black box and white box testing — answer in a 3-row table (see code?, what is tested?, who does it?).

CHECK YOURSELF

- Which phase produces the SRS document?
- Who usually performs white box testing?
- Name the three ways to deploy software.

Answers: 1. Requirement Analysis. 2. Developers. 3. Direct, Phased and Pilot release.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. SDLC is:

- A structured process from idea to retired software
- A logic gate
- A search algorithm
- A chart type

MCQ 2. The first SDLC phase is usually:

- Requirement analysis
- Maintenance
- Coding only
- Retirement

MCQ 3. A deliverable is:

- The document or product of a phase
- A Boolean 1
- A pie slice
- An IP port

MCQ 4. Testing comes:

- After implementation, before deployment
- Before requirements
- Instead of design
- Never

MCQ 5. Maintenance is:

- Fixing and updating after release
- The first interview
- NAND
- Big O

MCQ 6. SDLC exists to:

- Control quality, cost and time
- Replace Boolean algebra
- Draw K-maps
- Sort arrays

Extra short questions

- Define SDLC.
- List the six phases in order.
- Name one deliverable of requirements and of design.
- Why not jump straight to coding?

Extra long question

- Tabulate the six SDLC phases with activities and one deliverable each. ★ Golden 10-mark.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

- Software Development Life Cycle: a structured process that guides software from beginning to end.
- Requirements, design, implementation/coding, testing, deployment, maintenance.
- SRS / requirements document; design diagrams or architecture document.
- Wrong requirements become expensive bugs; SDLC catches them on paper first.

Long

- Phase | activities | deliverable. Stress that testing uses the requirements as the oracle, and maintenance is the longest phase in real life.



1.3.1 – 1.3.4 Communication Models & OSI 7-Layer Model ★ Golden

LEARN IT

A **communication model** explains how data travels from one device to another through a network. Because sending data is complicated, the job is split into **layers**; each layer does one part of the job and passes the data to the next layer.

The **OSI (Open Systems Interconnection) model** was developed by **ISO** (International Organization for Standardization) in **1984**. It divides communication into **7 layers**. When you send data, it goes **down** the layers on your device (7 → 1), travels as bits over the cable or air, and goes **up** the layers on the receiving device (1 → 7).



The 7 layers of the OSI model with the mnemonic "Please Do Not Throw Sausage Pizza Away".

REMEMBER IT

**Mnemonic (bottom to top, 1 → 7):** "Please Do Not Throw Sausage Pizza Away" = Physical, Data Link, Network, Transport, Session, Presentation, Application. (Top to bottom you can use "All People Seem To Need Data Processing".)

OSI layers — top layers (7 to 4)

| Layer           | Function                                                                          | Protocols / examples        |
|-----------------|-----------------------------------------------------------------------------------|-----------------------------|
| 7. Application  | Provides network services directly to user applications (browser, email program). | HTTP, HTTPS, FTP, SMTP, DNS |
| 6. Presentation | Data translation, encryption (security) and compression.                          | SSL/TLS, JPEG, MP3, ASCII   |
| 5. Session      | Opens, maintains and closes communication sessions.                               | NetBIOS, RPC                |
| 4. Transport    | Reliable end-to-end delivery, error checking, port numbering.                     | TCP (reliable), UDP (fast)  |

OSI layers — bottom layers (3 to 1)

| Layer        | Function                                                               | Protocols / hardware      |
|--------------|------------------------------------------------------------------------|---------------------------|
| 3. Network   | Logical addressing (IP addresses) and routing packets across networks. | IP, ICMP (Ping), Routers  |
| 2. Data Link | Physical addressing (MAC addresses) and error detection in frames.     | Ethernet, MAC, Switches   |
| 1. Physical  | Transmits raw bits (0, 1) over cables or wireless media.               | Cables, Hubs, Wi-Fi radio |

TCP vs UDP at the Transport layer

|              | TCP                                                             | UDP                                           |
|--------------|-----------------------------------------------------------------|-----------------------------------------------|
| Main feature | Reliable — checks that all data arrives correctly and in order. | Fast — sends data without waiting for checks. |
| Use when     | Accuracy matters.                                               | Speed matters more than perfection.           |
| Examples     | Web browsing, emails                                            | Video streaming, online gaming                |

WORKED EXAMPLE

Tracing an email through the OSI layers (sender side).

Layer 7 (Application): you press Send; your email program uses **SMTP**.

Layer 6 (Presentation): the text is encoded (ASCII) and may be encrypted with TLS.

Layer 5 (Session): a session is opened with the mail server.

Layer 4 (Transport): **TCP** breaks the data into segments and numbers them, so nothing is lost (email needs accuracy).

Layer 3 (Network): each packet gets the **IP address** of the destination; **routers** choose the path.

Layer 2 (Data Link): packets are put in **frames** with **MAC addresses**; a **switch** forwards them on the local network.

Layer 1 (Physical): the frames are sent as raw **bits** over the cable or Wi-Fi radio.

At the receiver, the same steps happen in reverse order (1 → 7).

COMMON MISTAKE

Do not mix up the two addresses: **IP address** = logical address = **Network layer (3)**; **MAC address** = physical address = **Data Link layer (2)**. Also, routers work at layer 3, switches at layer 2 and hubs at layer 1.

اردو میں سمجھیں

کیونیکیشن ماڈل بتاتا ہے کہ ڈیٹا نیٹ ورک کے ذریعے ایک ڈیوائس سے دوسری لک کیسے پہنچتا ہے۔ او ایس آئی (OSI) ماڈل 1984 میں آئی ایس او (ISO) نے بنایا اور اس کی سات ہیئیں (Layers) ہیں: فزیکل، ڈیٹا لنک، نیٹ ورک، ٹرانسپورٹ، سیشن، پریزنٹیشن اور ایپلی کیشن۔ ہر تپہ کا الگ کام ہے، جیسے نیٹ ورک تپہ IP ایڈریس اور روٹنگ سنبھالتی ہے۔ TCP درستگی کے لیے اور UDP رفتار کے لیے استعمال ہوتا ہے۔

EXAM TIP

A guaranteed long question: "Explain the seven layers of the OSI model." Draw the 7-layer stack (numbered, layer 7 on top), then write one function and one or two protocols/devices for each layer. MCQs often ask: "Which layer handles routing?" (Network), "Which layer does encryption?" (Presentation), "Who developed OSI and when?" (ISO, 1984), "Which protocol is used for video streaming?" (UDP).

CHECK YOURSELF

- Which OSI layer is responsible for encryption and compression?
- At which layer do switches and MAC addresses work?
- Should online gaming use TCP or UDP? Why?

Answers: 1. Presentation layer (6). 2. Data Link layer (2). 3. UDP, because speed matters more than perfect accuracy.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

|                                                                                     |                                                                                          |                                                                                                              |                                                                                                      |                                                                                                            |                                                                                                                                     |
|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>MCQ 1.</b> OSI has:<br>a. 7 layers<br>b. 4 layers<br>c. 2 layers<br>d. 12 layers | <b>MCQ 2.</b> Layer 7 is:<br>a. Application<br>b. Physical<br>c. Network<br>d. Data link | <b>MCQ 3.</b> IP addressing lives at:<br>a. Network (layer 3)<br>b. Physical<br>c. Application<br>d. Session | <b>MCQ 4.</b> Cables and bits are:<br>a. Physical<br>b. Presentation<br>c. Session<br>d. Application | <b>MCQ 5.</b> End-to-end ports (TCP/UDP) are:<br>a. Transport<br>b. Network<br>c. Physical<br>d. Data link | <b>MCQ 6.</b> The model that explains "how data travels" is a:<br>a. Communication model<br>b. Sort algorithm<br>c. K-map<br>d. MVP |
|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|

Extra short questions

- Write the 7 OSI layers from top to bottom.
- One job of Transport and of Network.
- Why layer models?
- Mnemonic: you will actually write in the exam.

Extra long question

- Name all 7 OSI layers with one job and one protocol/hardware example each. ★ Golden.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

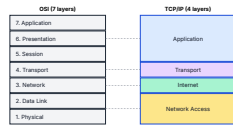
- Application, Presentation, Session, Transport, Network, Data Link, Physical.
- Transport: reliability/ports (TCP/UDP). Network: routing and logical addressing (IP).
- Each layer has a small job; vendors can change one layer without rewriting the rest.
- Please Do Not Throw Sausage Pizza Away (bottom-up) or All People Seem To Need Data Processing (top-down).

Long

- 7 App HTTP/DNS; 6 Pres encryption/JPEG; 5 Sess dialogs; 4 Trans TCP/UDP; 3 Net IP/routers; 2 DL frames/MAC/switches; 1Phy cables/bits/hubs. Draw the stack.

1.3.5 – 1.3.6 TCP/IP Model vs OSI Model

| LEARN IT                                                                                                                                                                                                                   |                                              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| The OSI model is mainly a <b>reference</b> model used for teaching and design. The <b>TCP/IP model</b> is the <b>practical model used on the real Internet</b> . It condenses the 7 OSI layers into just <b>4 layers</b> . |                                              |
| TCP/IP model (4 layers)                                                                                                                                                                                                    | Corresponds to OSI layers                    |
| 1. Application                                                                                                                                                                                                             | Application, Presentation, Session (7, 6, 5) |
| 2. Transport                                                                                                                                                                                                               | Transport (4)                                |
| 3. Internet                                                                                                                                                                                                                | Network (3)                                  |
| 4. Network Access                                                                                                                                                                                                          | Data Link, Physical (2, 1)                   |



How the 7 OSI layers map onto the 4 TCP/IP layers.

| Point            | OSI model                                             | TCP/IP model                                  |
|------------------|-------------------------------------------------------|-----------------------------------------------|
| Number of layers | 7                                                     | 4                                             |
| Use              | Reference / teaching model                            | Practical model used on the real Internet     |
| Top layers       | Separate Application, Presentation and Session layers | All three combined into one Application layer |
| Bottom layers    | Separate Data Link and Physical layers                | Combined into one Network Access layer        |

**WORKED EXAMPLE**

**Where does each protocol/device go in TCP/IP?**

HTTP (OSI layer 7) → TCP/IP **Application**.

SSL/TLS (OSI layer 6) → TCP/IP **Application** (layers 7, 6, 5 are merged).

UDP (OSI layer 4) → TCP/IP **Transport**.

IP and routers (OSI layer 3) → TCP/IP **Internet**.

Ethernet, switches, cables (OSI layers 2 and 1) → TCP/IP **Network Access**.

**REMEMBER IT**

Count it as **3-1-1-2**: the TCP/IP Application layer takes 3 OSI layers, Transport takes 1, Internet takes 1, Network Access takes 2. Total 3+1+1+2 = 7.

**COMMON MISTAKE**

The OSI "Network" layer (3) matches the TCP/IP "**Internet**" layer, not the TCP/IP "Network Access" layer. The similar names confuse many students.

آردو میں سمجھیں

ٹی سی پی/آئی پی (TCP/IP) ماڈل حقیقی انٹرنیٹ پر استعمال ہونے والا عملی ماڈل ہے اور اس کی صرف چار تہیں ہیں۔ اس کی ایپلی کیشن تہ OSI کی 7، 6 اور 5 تہوں کے برابر ہے، ٹرانسپورٹ تہ 4 کے، انٹرنیٹ تہ 3 کے، اور نیٹ ورک ایکسیس تہ 2 اور 1 کے برابر ہے۔

**CHECK YOURSELF**

1. How many layers does the TCP/IP model have?

2. Which TCP/IP layer corresponds to OSI layers 2 and 1?

3. Which TCP/IP layer contains the OSI Session layer?

Answers: 1. Four. 2. Network Access. 3. Application.

**Extra board practice**

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

**MCQ 1.** OSI has:

a. 7 layers

b. 4 layers

c. 2 layers

d. 12 layers

**MCQ 2.** Layer 7 is:

a. Application

b. Physical

c. Network

d. Data link

**MCQ 3.** IP addressing lives at:

a. Network (layer 3)

b. Application

c. Physical

d. Session

**MCQ 4.** Cables and bits are:

a. Physical

b. Presentation

c. Session

d. Application

**MCQ 5.** End-to-end ports (TCP/UDP) are:

a. Transport

b. Network

c. Physical

d. Data link

**MCQ 6.** The model that explains "how data travels" is a:

a. Communication model

b. Sort algorithm

c. K-map

d. MVP

**Extra short questions**

1. Write the 7 OSI layers from top to bottom.

2. One job of Transport and of Network.

3. Why layer models?

4. Mnemonic you will actually write in the exam.

**Extra long question**

1. Name all 7 OSI layers with one job and one protocol/hardware example each. ★ Golden.

**Extra answer key**

**MCQs:** 1. A 2. A 3. A 4. A 5. A 6. A

**Short**

1. Application, Presentation, Session, Transport, Network, Data Link, Physical.

2. Transport: reliability/ports (TCP/UDP). Network: routing and logical addressing (IP).

3. Each layer has a small job; vendors can change one layer without rewriting the rest.

4. Please Do Not Throw Sausage Pizza Away (bottom-up) or All People Seem To Need Data Processing (top-down).

**Long**

1. 7 App HTTP/DNS; 6 Pres encryption/JPEG; 5 Sess dialogs; 4 Trans TCP/UDP; 3 Net IP/routers; 2 DL frames/MAC/switches; 1 Phy cables/bits/hubs. Draw the stack.

# Computer Systems

12 single-page lessons · 7 Golden · read, close, say aloud, then sit the exam

## Chapter in one look

- Discrete quantities are counted (students, cars); continuous quantities are measured and can take any value (temperature, speed). Stairs vs ramp.
- A digital system uses only two values, 0 (OFF/LOW/FALSE) and 1 (ON/HIGH/TRUE), called bits. It is reliable, accurate, easy to design and programmable.
- Analog signals change smoothly (sine wave) and are affected by noise; digital signals jump between HIGH and LOW (square wave) and resist noise.
- Boolean algebra, by George Boole, works with only TRUE (1) and FALSE (0) using variables, operations and expressions.
- AND (·) gives 1 only when all inputs are 1; OR (+) gives 1 when any input is 1; NOT (') inverts its single input (unary).
- A truth table lists all  $2^n$  input combinations in binary order with their outputs.
- Logic gates are the building blocks of digital systems: AND, OR, NOT (basic); NAND, NOR (universal); XOR, XNOR (advanced).
- NAND and NOR are universal: any circuit can be built using only one of them.
- XOR gives 1 when inputs are different; XNOR gives 1 when inputs are the same.
- Order of precedence: Parentheses → NOT → AND → OR. A logic diagram is drawn in the same order.
- Minterms (SOP) come from rows with output 1; maxterms (POS) come from rows with output 0.
- K-maps use Gray code (00, 01, 11, 10); group 1s in sizes 1, 2, 4, 8, as large as possible; overlap and wrap-around are allowed; changing variables are eliminated.
- LogiSim Evolution simulates circuits: input pins are squares, output pins circles; the Poke Tool toggles inputs; bright green = 1, dark green = 0, blue = unconnected, red = error.
- SDLC has 6 phases: Requirement Analysis (SRS), Design, Implementation, Testing (black box/white box), Deployment (direct/phased/pilot), Maintenance.
- Waterfall is linear and suits fixed requirements (AI-Noor Library); Agile is iterative with 2–4 week sprints and suits changing requirements (AI-Noor Student App).
- The OSI model (ISO, 1984) has 7 layers — "Please Do Not Throw Sausage Pizza Away"; the practical TCP/IP model has 4 layers: Application, Transport, Internet, Network Access.

## Key terms

| Term                | Meaning                                                                   | اردو                       |
|---------------------|---------------------------------------------------------------------------|----------------------------|
| Discrete quantity   | A countable quantity with separate, fixed values.                         | منفصل مقدار                |
| Continuous quantity | A quantity that can take any value within a range.                        | مسلسل مقدار                |
| Bit                 | A binary digit, 0 or 1.                                                   | بائنری ہندسہ               |
| Digital system      | An electronic system that uses only the values 0 and 1.                   | ڈیجیٹل نظام                |
| Analog signal       | A continuous signal that changes smoothly, like a sine wave.              | اینالوگ سگنل (مسلسل اشارہ) |
| Digital signal      | A signal with only two levels, HIGH and LOW (square wave).                | ڈیجیٹل سگنل                |
| Noise               | Unwanted interference that disturbs a signal.                             | شور (مداخلت)               |
| Boolean algebra     | Algebra of two values, TRUE (1) and FALSE (0), developed by George Boole. | بولین الجبرا               |
| Truth table         | A table of all input combinations and their outputs.                      | تروٹھ ٹیبل (صداقت جدول)    |
| Logic gate          | An electronic circuit that performs a Boolean operation.                  | منطقی گیٹ                  |
| Universal gate      | A gate (NAND or NOR) from which any circuit can be built.                 | عالمگیر گیٹ                |
| Unary operation     | An operation with only one operand, e.g. NOT.                             | یک عددی عمل                |

# Computer Systems

75 marks · 1 hour 15 minutes · no notes · no phones

## CHAPTER EXAM RULES

**Instructions.** Sit this exam after Lesson 12 and the recap. **Section A** — all 15 MCQs (1 mark). **Section B** — all 10 short questions (3 marks). **Section C** — attempt any **three** long questions (10 marks). At least one Section C answer must be a ★ Golden topic. Write in your notebook. Do not open the key until time is up.

**Marks.**  $15 + 30 + 30 = 75$ . A 5-mark / 10-mark answer needs Define · Explain · Example · Diagram · Working.

سبق اور خلاصے کے بعد یہ امتحان بغیر نوٹس کے حل کریں۔ وقت پورا ہونے پر کلید کھولیں۔

## Section A — Multiple choice (15 × 1 = 15)

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1. Which of the following is a continuous quantity?
  - a. Number of students in a class
  - b. Number of cars in a parking lot
  - c. Temperature of water
  - d. Number of books on a shelf
2. In a digital system, the value 1 represents:
  - a. OFF
  - b. LOW
  - c. FALSE
  - d. ON / HIGH / TRUE
3. A square wave represents a:
  - a. Digital signal
  - b. Analog signal
  - c. Sine wave
  - d. Noise signal
4. Boolean algebra was developed by:
  - a. Charles Babbage
  - b. George Boole
  - c. Alan Turing
  - d. Maurice Karnaugh
5. How many rows does a truth table with 4 inputs have?
  - a. 4
  - b. 8
  - c. 16
  - d. 32
6. Which gate gives output 1 only when all inputs are 1?
  - a. OR
  - b. AND
  - c. NOR
  - d. XOR
7. Which pair of gates are called universal gates?
  - a. AND and OR
  - b. XOR and XNOR
  - c. NOT and AND
  - d. NAND and NOR
8. What is the output of a NOR gate when A = 0 and B = 0?
  - a. 1
  - b. 0
  - c. Unknown
  - d. Depends on the gate size
9. An XOR gate gives output 1 when:
  - a. Both inputs are 1
  - b. Both inputs are 0
  - c. The inputs are different
  - d. The inputs are the same
10. After parentheses, which operation has the highest precedence?
  - a. OR
  - b. NOT
  - c. AND
  - d. XOR
11. The minterm for the row A = 1, B = 0 is:
  - a. AB'
  - b. A'B
  - c. A + B'
  - d. A' + B
12. The columns of a 3-variable K-map (BC) are written in the order:
  - a. 00, 01, 10, 11
  - b. 00, 10, 01, 11
  - c. 11, 10, 01, 00
  - d. 00, 01, 11, 10
13. In black box testing, the tester:
  - a. Reads the source code line by line
  - b. Does not see the code and tests inputs/outputs
  - c. Is always the developer
  - d. Tests only the internal structure
14. Which OSI layer handles logical (IP) addressing and routing?
  - a. Data Link
  - b. Transport
  - c. Network
  - d. Session
15. The TCP/IP Network Access layer corresponds to which OSI layers?
  - a. 7, 6 and 5
  - b. 4
  - c. 3
  - d. 2 and 1

## Section B — Short questions (10 × 3 = 30)

---

1. Define discrete and continuous quantities. Give two examples of each.
2. State four advantages of digital systems.
3. Differentiate between analog and digital signals (any three points).
4. Why is NOT called a unary operation? Write its truth table.
5. How many rows are needed in a truth table for 3 inputs? List all the input combinations.
6. Why are NAND and NOR called universal gates? Show how a NAND gate can be used as a NOT gate.
7. A 2-input truth table has  $Y = 1$  only for rows  $A=0, B=1$  and  $A=1, B=0$ . Write the SOP expression and name the gate.
8. Name the deliverable of each SDLC phase.
9. Differentiate between black box testing and white box testing.
10. When should TCP be used and when should UDP be used? Give one example of each.

## Section C — Long questions (attempt any 3 × 10 = 30)

---

1. Draw the symbol, write the Boolean expression and give the truth table for each of these gates: AND, OR, NOT, NAND, NOR, XOR and XNOR.
2. A function is given as  $F(A, B, C) = \sum m(1, 3, 5, 6, 7)$ . (a) Make its truth table. (b) Simplify it using a 3-variable K-map, explaining each group. (c) Draw the logic diagram of the simplified expression.
3. For  $Y = A \cdot B' + C$ : (a) state the order in which the operations are done; (b) make the complete truth table; (c) draw the logic diagram.
4. Explain the six phases of the Software Development Life Cycle with the activities and deliverable of each phase. Use the house-building analogy.
5. Compare the Waterfall and Agile models under five headings. Explain why the AI-Noor Library Management System used Waterfall and the AI-Noor Student Mobile Application used Agile.
6. Explain the seven layers of the OSI model with the function and one protocol/device for each layer. Then show how they map onto the four layers of the TCP/IP model.

**Answer key.** Finish the Chapter Exam first, then mark. Do not peek while the clock is running.

پہلے امتحان مکمل کریں، پھر اس کلید سے نمبر لگائیں۔

## Check yourself — answers

1. **1.1.1 & 1.1.2 Discrete vs Continuous & Digital Systems.** Answers: 1. Discrete (you count pages). 2. Continuous (height is measured and can be any value). 3. The ramp.
2. **1.1.3 Analog and Digital Signals.** Answers: 1. A square wave. 2. Analog. 3. Human voice (or a traditional thermometer reading).
3. **1.1.4 – 1.1.6 Boolean Algebra, Operations & Truth Tables.** Answers: 1. George Boole. 2. 1 (because + means OR).
4. **1.1.7 Basic Logic Gates (AND, OR, NOT).** Answers: 1. 0. 2. Inverter. 3. Only  $A = 1, B = 1$ .
5. **1.1.7 Universal & Advanced Logic Gates.** Answers: 1. Any logic circuit can be built using only NAND gates or only NOR gates. 2. 0. 3. Join both inputs together and feed A into them:  $(A \cdot A)' = A'$ .
6. **1.1.8 – 1.1.11 Logic Expressions, Minterms & Diagrams.** Answers: 1. AND. 2.  $AB'C$ . 3.  $A + B'$ .
7. **1.1.13 Karnaugh Maps (K-Maps) Simplification.** Answers: 1.  $BC = 11$ . 2. Cells 2 and 3 are the row  $A = 1$ ; B changes  $\rightarrow Y = A$ . 3. The whole row  $A = 1$  is a group of 4  $\rightarrow Y = A$ .
8. **LogiSim Simulation Tool Guide.** Answers: 1. In the Attribute Table (bottom left). 2. An error or short circuit.
9. **1.2.1 – 1.2.3 Software Development Life Cycle (SDLC).** Answers: 1. Requirement Analysis. 2. Developers. 3. Direct, Phased and Pilot release.
10. **1.2.4 – 1.2.7 Waterfall vs Agile Models & Case Studies.** Answers: 1. Because each phase must be completed before the next begins, in a fixed order. 2. Because its requirements were 100% fixed before coding began.
11. **1.3.1 – 1.3.4 Communication Models & OSI 7-Layer Model.** Answers: 1. Presentation layer (6). 2. Data Link layer (2). 3. UDP, because speed matters more than perfect accuracy.
12. **1.3.5 – 1.3.6 TCP/IP Model vs OSI Model.** Answers: 1. Four. 2. Network Access. 3. Application.

## Answer key

### Multiple choice

1. c
2. d
3. a
4. b
5. c
6. b
7. d
8. a
9. c
10. b
11. a
12. d
13. b
14. c
15. d

## Short questions — model answers

1. Discrete quantities are separate, countable items with specific values (number of students in a classroom, number of cars in a parking lot). Continuous quantities can take any value within a range and flow without breaks (temperature of water, speed of a moving car).
2. Reliable (less affected by noise and interference); accurate (no loss of quality when copying); easy to design (only two states, 0 and 1); programmable (easily controlled by software).
3. Nature: analog is a continuous flow; digital has discrete values (bits, HIGH/LOW). Shape: analog is a sine wave; digital is a square wave. Reliability: analog is low (affected by noise); digital is high (resistant to noise). Examples: human voice vs computer data.
4. NOT works on only one operand (one input), so it is unary. Truth table:  $A = 0 \rightarrow Y = 1$ ;  $A = 1 \rightarrow Y = 0$ .
5.  $2^3 = 8$  rows: 000, 001, 010, 011, 100, 101, 110, 111.
6. Any logic circuit can be built using only NAND gates, or only NOR gates. NAND as NOT: join both inputs and feed A:  $Y = (A \cdot A)' = A'$ . Check:  $A = 0 \rightarrow 1$ ,  $A = 1 \rightarrow 0$ .
7. Minterms:  $A'B$  (row 01) and  $AB'$  (row 10).  $Y = A'B + AB'$ . This is the XOR gate,  $Y = A \oplus B$ .
8. Requirement Analysis  $\rightarrow$  SRS document; System Design  $\rightarrow$  Design Document; Implementation  $\rightarrow$  Working Source Code; Testing  $\rightarrow$  Test Reports / bug-free software; Deployment  $\rightarrow$  Live Software; Maintenance  $\rightarrow$  Patches / Updated Versions.
9. Black box: tester does not see the code, tests functionality (inputs/outputs), done by dedicated testers. White box: tester can see the code, tests internal logic/structure, done by developers.
10. Use TCP when accuracy matters, e.g. web browsing or email. Use UDP when speed matters more than perfection, e.g. video streaming or online gaming.

## Long questions — model answers

1. Symbols as drawn in 1.1.7 (AND = D shape; OR = curved back, pointed front; NOT = triangle + bubble; NAND = AND + bubble; NOR = OR + bubble; XOR = OR with extra back curve; XNOR = XOR + bubble). Expressions: AND  $Y = A \cdot B$ ; OR  $Y = A + B$ ; NOT  $Y = A'$ ; NAND  $Y = (A \cdot B)'$ ; NOR  $Y = (A + B)'$ ; XOR  $Y = A \oplus B$ ; XNOR  $Y = (A \oplus B)'$ . Outputs for inputs 00, 01, 10, 11: AND 0,0,0,1; OR 0,1,1,1; NAND 1,1,1,0; NOR 1,0,0,0; XOR 0,1,1,0; XNOR 1,0,0,1. NOT: 0  $\rightarrow$  1, 1  $\rightarrow$  0.
2. (a) Truth table ( $ABC \rightarrow F$ ): 000 $\rightarrow$ 0, 001 $\rightarrow$ 1, 010 $\rightarrow$ 0, 011 $\rightarrow$ 1, 100 $\rightarrow$ 0, 101 $\rightarrow$ 1, 110 $\rightarrow$ 1, 111 $\rightarrow$ 1.

(b) K-map:

|       | BC = 00 | BC = 01 | BC = 11 | BC = 10 |
|-------|---------|---------|---------|---------|
| A = 0 | 0       | 1       | 1       | 0       |
| A = 1 | 0       | 1       | 1       | 1       |

Group 1 (size 4): the two middle columns,  $m_1, m_3, m_5, m_7$ . A changes and B changes; C = 1 stays  $\rightarrow$  **C**. Group 2 (size 2):  $m_7$  and  $m_6$  in row A = 1. C changes; A = 1, B = 1 stay  $\rightarrow$  **AB** ( $m_7$  is shared — overlapping is allowed). **F = C + AB**. Check: C + AB is 1 for rows 1, 3, 5, 7 (C = 1) and 6, 7 (AB = 1) = {1, 3, 5, 6, 7}. ✓

(c) Logic diagram: an AND gate with inputs A and B gives AB; an OR gate takes AB and C and gives F.

3. (a) First NOT ( $B'$ ), then AND ( $A \cdot B'$ ), then OR ( $+ C$ ).

(b) Truth table:

| A | B | C | B' | A·B' | Y |
|---|---|---|----|------|---|
| 0 | 0 | 0 | 1  | 0    | 0 |
| 0 | 0 | 1 | 1  | 0    | 1 |
| 0 | 1 | 0 | 0  | 0    | 0 |
| 0 | 1 | 1 | 0  | 0    | 1 |
| 1 | 0 | 0 | 1  | 1    | 1 |
| 1 | 0 | 1 | 1  | 1    | 1 |
| 1 | 1 | 0 | 0  | 0    | 0 |
| 1 | 1 | 1 | 0  | 0    | 1 |

(c) Logic diagram: input B goes through a NOT gate to give  $B'$ . A and  $B'$  go into an AND gate to give  $A \cdot B'$ .  $A \cdot B'$  and C go into an OR gate to give Y. (3 gates.)

4. Requirement Analysis: meet clients, understand needs, list features; most important phase; deliverable SRS (house: asking the family what rooms they need). System Design: high-level design (architecture), low-level design (database, UI screens), flowcharts; deliverable Design Document (architect draws blueprints). Implementation: write code in Python, Java etc. from the design; deliverable Working Source Code (builders lay bricks). Testing: find and fix bugs using black box and white box testing; deliverable Test Reports / bug-free software (inspector checks the house). Deployment: release to end-users by direct, phased or pilot release; deliverable Live Software (family moves in). Maintenance: support, updates, fixing post-release issues; deliverable Patches / Updated Versions (repairs and extensions).
5. Comparison: Flexibility — Waterfall rigid, Agile very flexible. Customer involvement — Waterfall only at start and end, Agile continuous after every sprint. Delivery — Waterfall single delivery at the end, Agile frequent partial deliveries. Testing — Waterfall late in a dedicated phase, Agile continuous in every sprint. Documentation — Waterfall heavy, Agile light. The Al-Noor Library system used Waterfall because its requirements were 100% fixed before coding began, so a simple, well-documented linear process fitted. The Al-Noor Student App used Agile because teachers, parents and students kept requesting new features, so short sprints with continuous feedback were needed.
6. OSI (ISO, 1984), top to bottom: 7 Application — network services to user applications (HTTP, HTTPS, FTP, SMTP, DNS); 6 Presentation — translation, encryption, compression (SSL/TLS, JPEG, MP3, ASCII); 5 Session — opens, maintains, closes sessions (NetBIOS, RPC); 4 Transport — reliable end-to-end delivery, error checking, port numbers (TCP, UDP); 3 Network — logical IP addressing and routing (IP, ICMP, routers); 2 Data Link — physical MAC addressing and error detection in frames (Ethernet, switches); 1 Physical — raw bits over cables/wireless (cables, hubs, Wi-Fi radio). Mnemonic bottom to top: "Please Do Not Throw Sausage Pizza Away". TCP/IP mapping: Application = OSI 7, 6, 5; Transport = 4; Internet = 3; Network Access = 2, 1.

## Self-assessment: I can...

- ☐ ...tell discrete from continuous quantities and analog from digital signals, and draw both waves.
- ☐ ...evaluate AND, OR and NOT expressions and follow the order of precedence.
- ☐ ...draw all seven gate symbols and write their truth tables from memory.
- ☐ ...write the SOP (minterms) and POS (maxterms) from a truth table.
- ☐ ...simplify expressions with 2- and 3-variable K-maps, including overlap and wrap-around groups.
- ☐ ...explain the six SDLC phases and choose between Waterfall and Agile for a project.
- ☐ ...list four advantages of digital systems.
- ☐ ...build a complete truth table for 2 or 3 inputs without missing a row.
- ☐ ...explain why NAND and NOR are universal gates.
- ☐ ...draw a logic diagram from an expression.
- ☐ ...build and test a simple circuit in LogiSim and read the wire colours.
- ☐ ...name the 7 OSI layers in order, state their functions, and map them to the 4 TCP/IP layers.